Q2.)

Part A.) $\frac{d^2x}{dt^2} = -GM\frac{x}{r^3}$

$$\frac{d^2y}{dt^2} = -GM\frac{y}{r^3}$$

$X_1 = x$, $X_2 = y$

$X_3 = \dot{x} = V_x$, $X_4 = \dot{y} = V_y$

$r = \sqrt{X_1^2 + X_2^2}$

$\dot{X}_1 = X_3 = V_x$

$\dot{X}_2 = X_4 = V_y$

$\dot{X}_3 = \frac{-GMX_1}{(X_1^2+X_2^2)^{3/2}}$, $\dot{X}_4 = \frac{GMX_2}{(X_1^2+X_2^2)^{3/2}}$

$\dot{x} = V_x$, $\dot{y} = V_y$

$\dot{V}_x = \frac{-GMx}{(x^2+y^2)^{3/2}}$, $\dot{V}_y = \frac{-GMy}{(x^2+y^2)^{3/2}}$